Misinterpretation Insertion for Charts & Function

You are given a math problem with a **coordinate axis or chart** and its solution. Your task is to introduce errors into the solution by misreading the image or chart and generate an incorrect solution.
Instructions:

- Your response involves three Stages:
  ** Stage 1: Image Information Extraction **: Identify where in the solution diagram information is extracted. Note these areas as potential points for introducing misinterpretations.
  ** Stage 2: Introduce a Misinterpretation **: Choose one of the identified points from Stage 1 and alter it to create a misreading scenario. Integrate this misinterpretation naturally into the solution.
  ** Stage 3: Continue Reasoning **: Based on the misinterpretation from Stage 2, continue the reasoning process to derive an incorrect final answer.

- Ensure the misreading is naturally integrated without explicit statements about making a misinterpretation. When describing corrupted solution, avoid showing the misleading thought process. You must respond in a typical problem-solving style.
- Remember that the <pos> and <neg> tags only need to appear at the end of each Step X. Do not repeat them between Step X and Step X+1, even if the reasoning process for that step spans many lines. Remember that once a step is marked as <neg>, all subsequent steps are also considered <neg>.
- '†Answer:' is the final answer, and it should not be tagged.
- You cannot tag an unaltered step with <neg> without misinterpreting it. You must misinterpret the step if you are tagging it with <neg>.
- Words that imply misinterpretation, such as ‘misread’ ‘incorrect’ or ‘incorrectly assume' must not appear in your response under any circumstances.
- The misinterpretation action in Stage 3 should be consistent with what was planned in Stage 2.

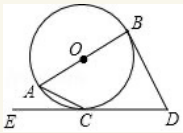
Question:
{QuestionText}

Solution:
{SolutionText}

** Your response:

Figure 17: Misinterpretation insertion for function and chart-related problems.

URSA-RM-8B as Verifier



Problem:
As shown in the figure, AB is the diameter of circle O, DB and DC are respectively tangent to circle O at points B and C. If ∠ACE = 25.0°, then the degree of ∠D is ()
A. 50° B. 55° C. 60° D. 65°

Solution 1:
Step 1: We are given that AB is the diameter of circle O, and DB and DC are tangents to circle O at points B and C respectively. We are also given that ∠ACE = 25°. 0.93

Step 2: Since DB and DC are tangents to circle O from the same point D, we know that DB = DC. This means that triangle DBC is an isosceles triangle. 0.95

Step 3: In triangle DBC, let ∠DBC = ∠DCB = x. Since the sum of angles in a triangle is 180°, we have ∠BDC + ∠DBC + ∠DCB = 180°. Therefore, ∠BDC + 2x = 180°. 0.91

Step 4: Since AB is the diameter, ∠ACB is a right angle (90°). In right-angled triangle ACB, we have ∠CAB + ∠ABC = 90°. 0.93

Step 5: We are given that ∠ACE = 25°. Since ∠ACB = 90°, we have ∠ECB = 90° - 25° = 65°. 0.98

Step 6: In triangle DBC, ∠DBC + ∠DCB + ∠BDC = 180°. Since DB and DC are tangents, ∠DBC = ∠DCB = 90° - 25° = 65°. Therefore, 65° + 65° + ∠BDC = 180°, which gives ∠BDC = 180° - 130° = 50°. 0.98

Step 7: Therefore, ∠D = ∠BDC = 50°. 0.88

†Answer: A ✓

Solution 2:
Step 1: Identify the relevant angles. We are given that ∠ACE = 25°. We want to find ∠D. 0.99

Step 2: Recognize the relationship between the angles. Since AB is the diameter of circle O, and DB is tangent to circle O at B, we know that ∠ABD is a right angle (90°). Also, since DB and DC are tangents from the same point D to circle O, we know that DB = DC. This means that triangle DBC is an isosceles triangle. 0.94

Step 3: Use the property of isosceles triangles. In isosceles triangle DBC, ∠DBC = ∠DCB. Let's denote this angle as x. The sum of angles in triangle DBC is 180°. Therefore, ∠BDC + ∠DBC + ∠DCB = 180°. Substituting the values, we get ∠BDC + x + x = 180°, which simplifies to ∠BDC + 2x = 180°. 0.90

Step 4: Relate the angles. ∠ACE is an inscribed angle subtending the arc BC. The angle subtended by the same arc at the center is twice the inscribed angle. Therefore, ∠BOC = 2 * ∠ACE = 2 * 25° = 50°. 0.46

Step 5: Find the angle ∠BDC. Since ∠ABD is a right angle (90°), and ∠BOC is 50°, we can find ∠BDC. In quadrilateral ABCD, the sum of angles is 360°. We have ∠DAB + ∠ABC + ∠BCD + ∠CDA = 360°. Since ∠ABC = 90° and ∠BCD = x, we have ∠DAB + 90° + x + ∠CDA = 360°. 0.67

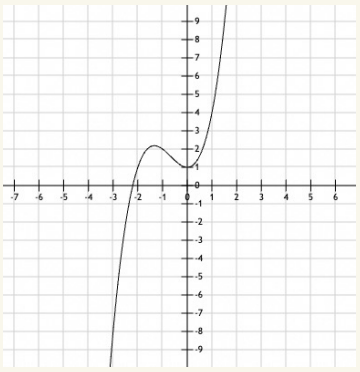
Step 6: Solve for x. We know that ∠BDC + 2x = 180°. Also, ∠BDC + ∠DAB = 180° (angles on a straight line). Therefore, ∠DAB = 180° - ∠BDC. Substituting this into the equation from step 5, we get (180° - ∠BDC) + 90° + x + ∠CDA = 360°. This simplifies to 270° - ∠BDC + x + ∠CDA = 360°. 0.65

Step 7: Determine ∠D. Since triangle DBC is isosceles, ∠DBC = ∠DCB = x. Also, ∠BDC + 2x = 180°. We know that ∠ACE = 25°, which is an inscribed angle subtending arc BC. The central angle subtending the same arc is 2 * 25° = 50°. This central angle is ∠BOC. In triangle OBC, OB = OC (radii), so it's an isosceles triangle. Therefore, ∠OBC = ∠OCB = (180° - 50°)/2 = 65°. Since ∠OBC and ∠DBC are supplementary, ∠DBC = 180° - 65° = 115°. However, this is incorrect. Let's reconsider. Since ∠ACE = 25°, the arc BC subtends an angle of 50° at the center. In triangle OBC, OB = OC, so ∠OBC = ∠OCB = (180° - 50°)/2 = 65°. Therefore, ∠D = 65°. 0.47

†Answer: D ✗

Figure 18: Case of URSA-8B-RM serving as a verifier in Best-of-N evaluation.

Showcase of how MIE works



Problem:
Given a cubic function y=ax^3+bx^2+cx+d, please determine the values of a, b, c and d.

Original Solution:
Step 1: I first need to identify the points that the graph passes through, and then substitute these points into the equation to solve for the values of a, b, c, and d.
Step2: I find some points below: P1: (-3, -8) P2: (0, 1) P3: (-2, 1) P4: (-1, 2)
…
Step N: a=1, b=2, c=0, d=1

Gemini's response:
Information Extraction: I see this question is highly related to the coordinates of points on the line. The line appears in the third quadrant, second quadrant, and first quadrant.

Misinterpretation Insertion: A simple method is to introduce incorrect point coordinates to reach a wrong conclusion. As the correct solution states, the coordinates of point P1 are (-3, -8), but we easily replace them with (-3, -7) as a premise.

Continue Reasoning:
Step 1: I first need to identify the points that the graph passes through, and then substitute these points into the equation to solve for the values of a, b, c, and d. <pos>
Step 2: I find some points below: P1: (-3, -7) P2: (0, 1) P3: (-2, 1) P4: (-1, 2). <neg>
Step 3: Using P1: (-3, -7),
−7=a(−3)^3+b(−3)^2+c(−3)+1
−7=−27a+9b−3c+1
-8= −27a+9b−3c <neg>
…
Step N: $a = \frac{5}{6}, b = \frac{3}{2}, c = -\frac{1}{3}, d = 1$. <neg>

Figure 19: Case from MIE. We introduce specific step-level perception errors and continue false reasoning to construct a correctnesses-labeled solution.